

Employee Attrition Prediction – HR Summary

Project Objective

This project analyzed employee data to identify patterns associated with employee attrition and to help the HR department proactively retain valuable employees.

Key Findings

Approximately 16% of employees in the dataset left the organization. The analysis showed that overtime, certain job roles (especially Laboratory Technician and Sales Representative), frequent business travel, and overall experience were among the strongest indicators of attrition.

Business Impact

Employees in high-risk roles and departments should receive additional attention through regular check-ins, career development opportunities, and workload management. Addressing these issues can reduce recruitment costs and improve employee retention.

Recommendations

1. Monitor employees who frequently work overtime and provide better work-life balance.
2. Focus retention efforts on high-risk departments and job roles.
3. Conduct career development and promotion discussions with employees showing higher attrition risk.
4. Use the prediction model as a decision-support tool alongside HR judgment.

Model Limitation

The model is based on historical HR data and cannot capture every personal or organizational reason why an employee may resign. Predictions should support—not replace—human decision-making.